Making HEPA's bound empirical: an a priori test for whether HEPA will work on a new dataset

Aviral Garg · Forgis research proposal · 10 October 2026 · on HEPA (Petersen et al., arXiv:2605.11130v4)

The gap I want to close, stated by the paper itself

HEPA's Proposition 1 gives $I(H_t; E_{t+\Delta t}) \geq I(H^*; E_{t+\Delta t}) - C_\eta L^2 \varepsilon$, with vacuity threshold (their eq. 17) $\varepsilon_{\text{vac}} = I(H^*; E_{t+\Delta t})/(C_\eta L^2)$.

- Within a dataset the bound is validated: varying ε via pretraining snapshots gives Spearman $\rho = -0.87$ (C-MAPSS-1), -0.67 (C-MAPSS-3), -0.64 (MBA), -0.49 (SMAP).
- Across datasets it falls apart. Pooling converged ε over the 14 benchmarks gives Pearson $r = -0.05$ ($p = 0.90$). The paper attributes this to L, C_η and target scale differing by dataset.
- The constants are never estimated. In the authors' words: "The constants L and C_η are not estimated directly; fig. 3 validates only the monotonic relationship, not the full quantitative bound," and in Future Work: "deriving fully empirical versions of the information-retention bound that estimate L and C_η directly from data remains an interesting open problem."

As it stands the bound only explains HEPA's failures after the fact. I want to make it predictive. Define the dimensionless slack

$$\hat{\rho}(\mathcal{D}) \;=\; \frac{\hat{\varepsilon}}{\hat{\varepsilon}_{\text{vac}}} \;=\; \frac{\hat{\varepsilon} \cdot \hat{C}_\eta \hat{L}^2}{\hat{I}(H^*; E)}, \qquad \text{bound non-vacuous} \iff \hat{\rho} < 1.$$

Hypotheses

- H1 (normalisation restores the cross-dataset signal). Across the 14 benchmarks, $\rho_s(\hat{\rho},\ \text{h-AUROC}) \leq -0.6$, where raw ε gives $r = -0.05$. Falsified if $|\rho_s| < 0.3$, which would mean the incommensurability comes from somewhere else and my normalisation is wrong. The objection I have to design against: $\hat{I}(H^*; E)$ comes from a probe fitted on the same labels h-AUROC uses, so part of any correlation could be the probe predicting itself. Control: report the partial correlation of $\hat{\rho}$ with h-AUROC given the probe's own AUROC, on disjoint rows. If it vanishes there, H1 is dead and I would say so.
- H2 (it works as a screen). A prediction: I have not computed $\hat{\rho}$ anywhere yet. $\hat{\rho} \geq 1$ should mark where the bound gives no guarantee, from unlabelled pretraining plus a small probe. Pre-registered control: CHB-MIT, excluded at h-AUROC 0.497 with $I(H^*; E) \approx 0$ (appendix A.6), should land at $\hat{\rho} \geq 1$; C-MAPSS-1 (0.806) at $\hat{\rho} \ll 1$. Falsified if either is misclassified. What it does not mean: appendix A.4 says a vacuous bound does not imply $I(H_t; E) = 0$, and GECCO proves it: within-dataset $\rho = +0.14$ ($p = 0.67$), yet HEPA scores 0.88 there, its best in table 1. The screen is about the guarantee, not about failure.
- H3 (the L1/L2 gap costs more than monotonicity). A2 is in L2 while eq. (2) trains L1. Appendix A.2 already answers this via $\|u\|_2 \leq \|u\|_1$, calling L1 a "monotone proxy for ε", which is enough for the within-dataset test in fig. 3 but not here, since $\hat{\rho} \geq 1$ needs ε on its scale rather than its ordering. Prediction: $\hat{\rho}$ from the measured L2 residual beats the L1 version by $\Delta\rho_s \geq 0.15$. Falsified if they sit within noise, which would license the L1 loss.

Estimators, all cheap and low-dimensional where it matters

Fit HEPA's own event head $\hat{\eta}(h) = \sigma(w^\top \text{LN}(h) + b)$ (LayerNorm + linear, 769 params on d=256, their appendix C) on the frozen target representations h^*. Then, on held-out data:

- $\hat{L} = \|w\|_2/4$. For this head the Lipschitz constant is exact, not estimated: $\nabla_h \hat{\eta} = \sigma'(z)\, w$ and $\sigma' \leq 1/4$. A kNN finite-difference quantile is then only a sanity check on the bound's tightness, not the estimator.
- $\hat{C}_\eta = \big(2\,\underline{\eta}\,(1 - \overline{\eta})\big)^{-1}$ with $\underline{\eta}, \overline{\eta}$ the 1st/99th percentiles of $\hat{\eta}(h^*)$.
- $\hat{I}(H^*; E) = H(\pi_e) - \mathbb{E}\big[H(\hat{\eta}(H^*))\big]$, plug-in. E is binary, so this is a 1-D estimate. No high-dimensional MI estimator is needed, which is usually where this kind of proposal dies.
- $\hat{\varepsilon} = \mathbb{E}\|\hat{h} - h^*\|_2^2$ on a held-out split, and in L1 as well, per H3.

Why it is worth doing, and what it costs

- Scientific: it makes HEPA's central theorem computable and falsifiable, answering the open problem the paper names itself.
- Industrial: $\hat{\rho}$ needs unlabelled pretraining and a small probe, so you can tell before committing whether the guarantee holds on a new plant.
- Cost: the public repo ships data-download and training scripts, so this is a re-run of HEPA's recipe, not a new model. Pretraining is under a minute per dataset on one A10G; the estimators are $O(N)$.

Risks I can already name

- Plug-in MI biases $\hat{I}$ down, pushing $\hat{\rho}$ up toward the threshold. Rank correlation protects H1; H2 cannot hide behind that, being an absolute test at $\hat{\rho} = 1$, so I would calibrate the threshold on held-out folds and report the bias direction with every call.
- $n = 14$ datasets is low power. Pre-register the direction and the test before computing any correlation, and show the per-dataset scatter, not just the coefficient.

Secondary direction (from my analysis of `embeddings.npz`)

In the attached report, with a frozen target no SIGReg weight $\alpha \in [0.01, 0.9]$ bought both prediction quality and representation rank (at α=0.1: cosine 0.8716±0.0004→0.0609±0.0044, 5 seeds). HEPA does not hit this: its target is never frozen (weight-shared by default, and robust to an EMA target per appendix I.3, which chooses joint training "for its simplicity"), so it stays free to migrate toward the isotropic Gaussian SIGReg wants. The frozen-target regime sits outside both settings I.3 tests. SIGReg is doing two jobs at once: preventing collapse, and setting the target's distribution.

- H4. Split the jobs: SIGReg on the target branch, an explicit rank floor on the predictor output. Prediction: h-AUROC goes flat in α over $[0.03, 0.3]$ (range ≤ 0.02), with predictor-output effective rank as the mediator. Falsified if h-AUROC still varies by more than 0.02 there, or if the rank floor binds without moving h-AUROC. This is not a claim about the BATADAL/MBA gap, which the paper attributes to tokenisation and scopes separately.

Worth confirming: the paper reports α=0.1 (table 7) while `Forgis-Labs/HEPA` PR #3 says the table-1 runs used 0.04. I've worked from 0.1.